Network Traffic Timeline Analysis Report

Milestone 3: Detection & Visualization

Analysis dataset: Combined_Traffic_Data.csv

Source: Combined PCAP-derived dataset from three Milestone 2 traffic segments

Total records analysed: 3,588

Analysis type: Timestamp-based network traffic timeline analysis

Visualization: Network Traffic Timeline Analysis

1. Objective

The purpose of this analysis was to examine the temporal distribution of network traffic across the combined PCAP dataset and identify significant changes in traffic volume over the duration of the capture.

The analysis was performed as part of Milestone 3: Detection & Visualization, which requires investigation of network behaviour, identification of anomalies or significant traffic patterns, use of visualization techniques, AI-assisted interpretation, and verification of analytical results.

The timeline analysis specifically focused on:

- Traffic volume over time
- Identification of high-volume and low-volume periods
- Identification of significant traffic bursts or changes in activity
- Verification of the capture duration and timestamp coverage
- Verification of complete timestamp coverage for the combined dataset
- Interpretation of temporal behaviour without incorrectly treating high traffic volume as proof of malicious activity

2. Source Data Preparation

The network investigation originally consisted of three separate PCAP traffic segments analysed during Milestone 2. Rather than analysing the segments independently for the final visualization, the three extracted datasets were combined into a single consolidated dataset.

A Python-based data preparation process was used to load the traffic data from the three Milestone 2 segments, combine the records into a single dataset, preserve the relevant packet metadata fields, verify the resulting record count, and generate the consolidated source file used for subsequent analysis.

Output file: Combined_Traffic_Data.csv

The final combined dataset contains 3,588 total records, made up of 3,549 IP-bearing records and 39 ARP/non-IP records. Timestamp information was available for all 3,588 captured records used in the timeline analysis.

Evidence

The Python preparation process is supported by two screenshots already saved in the Milestone 3 project folder.

Screenshot 1: Python Library Verification

Figure: Python environment/library verification

This screenshot documents the Python preparation and verification stage performed before the traffic datasets were combined.

Screenshot 2: Combined Dataset Creation

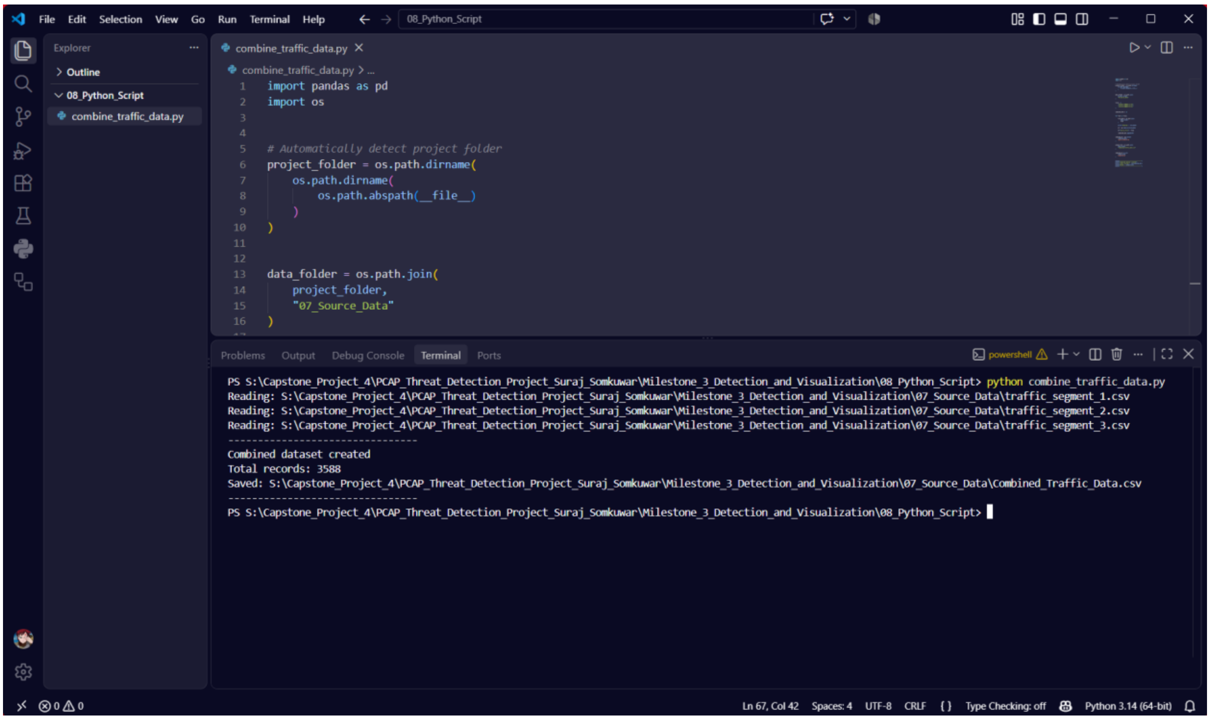


Figure: Python script used to combine the three traffic segments

This screenshot provides evidence that Python was used to combine and verify the dataset before the subsequent analysis and visualization stages.

3. Timeline Analysis Methodology

The timeline analysis was performed using the frame.time field contained in Combined_Traffic_Data.csv.

An important characteristic of the source dataset was identified during verification: the physical order of rows in the CSV is not necessarily chronological. For example, records occurring around 02:06, 02:07, and 02:08 may appear in the file before records occurring around 02:05. Therefore, the physical row order was not used to determine the temporal sequence.

Instead, the analysis followed a timestamp-based methodology:

1. Read the frame.time value for each record.
2. Validate timestamp values and convert them into datetime values.
3. Include timestamp-valid records in chronological time binning.
4. Assign each record to its corresponding one-minute interval based on its actual timestamp.
5. Aggregate the number of records in each one-minute interval.
6. Arrange the resulting intervals chronologically for visualization.

This approach ensures that a record is counted according to when it actually occurred, rather than where it happens to appear in the CSV file.

4. Capture Period Verification

The actual timestamp values in the combined dataset establish the capture period.

Capture start: 2021-08-17 01:58:06.017228 +0530

Capture end: 2021-08-17 02:08:16.417023 +0530

Capture duration: Approximately 10 minutes 10.40 seconds (10.1733 minutes)

The combined dataset contains 3,588 records, and timestamp information is available for all 3,588 records used in the timeline analysis. All records were included in the chronological one-minute aggregation.

5. Verified One-Minute Traffic Distribution

The actual Combined_Traffic_Data.csv was programmatically checked against the timeline visualization. The resulting one-minute distribution is shown below.

Capture Interval	Records
01:58 to 01:59	696
01:59 to 02:00	2,097
02:00 to 02:01	54
02:01 to 02:02	82
02:02 to 02:03	113
02:03 to 02:04	68
02:04 to 02:05	104
02:05 to 02:06	81
02:06 to 02:07	103

Capture Interval	Records
02:07 to 02:08	150
02:08 to 02:09 (partial)	40
Total timestamp-valid records	**3,588**

The final interval is partial because the capture ended at 02:08:16.417023 +0530, rather than continuing until 02:09:00.

Independent Verification Result

The 11 calculated interval counts were compared directly against the values displayed in the final timeline visualization. All 11 interval values matched exactly, and their combined total reconciles to the 3,588 records in the dataset. This confirms that the timeline visualization is supported by the underlying Combined_Traffic_Data.csv dataset.

6. Important Timeline Findings

6.1 Peak Traffic Period

The most significant temporal observation is the traffic burst during the 01:59 to 02:00 interval, which contains 2,097 records: the highest one-minute traffic volume observed in the combined dataset. Relative to the complete dataset, this works out to 2,097 / 3,588 x 100 = 58.44%. Approximately 58.44% of all captured records therefore occurred during this single one-minute interval, representing a substantial concentration of traffic within a short period.

6.2 Traffic Reduction After the Peak

Immediately after the peak interval, traffic volume decreased sharply: from 2,097 records during 01:59 to 02:00, down to 54 records during 02:00 to 02:01, a difference of 2,043 fewer records. This demonstrates a pronounced reduction in observed traffic volume immediately following the peak. The subsequent intervals contain considerably lower volumes, ranging between 68 and 150 records for most of the later full-minute intervals.

6.3 Later Traffic Activity

After the large 01:59 to 02:00 burst, the observed traffic volume remained comparatively low across the later full-minute intervals: 82 records from 02:01 to 02:02, 113 records from 02:02 to 02:03, 68 records from 02:03 to 02:04, 104 records from 02:04 to 02:05, 81 records from 02:05 to 02:06, 103 records from 02:06 to 02:07, and 150 records from 02:07 to 02:08.

The highest later full-minute interval was 02:07 to 02:08, with 150 records. Compared with the peak of 2,097 records, this demonstrates that the exceptionally high traffic concentration was primarily associated with the 01:59 to 02:00 interval.

7. Lowest and Partial Traffic Interval

The final interval, 02:08 to 02:09, contains 40 records: the lowest observed interval count in the dataset. However, it should not be directly compared with the complete one-minute intervals without qualification, because the capture ended at 02:08:16.417023, meaning only approximately the first 16 seconds of that interval are represented. The timeline visualization correctly marks this interval with an asterisk to indicate that it is partial.

8. Statistical Summary

The verified timestamp-based timeline contains 11 one-minute intervals representing the complete capture period, with the final interval being partial.

Statistic	Value
Total timestamp-valid records	3,588
Maximum one-minute count	2,097
Minimum observed interval count	40
Mean interval count	approximately 326.18 records
Median interval count	103 records

The large difference between the mean and median is influenced by the exceptional 2,097-record peak interval.

The complete capture rate can also be calculated using the total dataset: 3,588 records divided by 10.1733 minutes gives approximately 352.69 records per minute. This represents the average record rate over the complete capture duration.

All 3,588 records in the combined dataset were included in the chronological timeline analysis.

9. AI-Assisted Analysis and Interpretation

AI tools were used during Milestone 3 to assist with analysis and interpretation of the combined traffic dataset and the resulting visualizations.

For the timeline analysis, AI-assisted processing was used to identify the overall traffic-volume pattern, the highest traffic interval, the lowest observed interval, changes in traffic volume following the peak, the capture duration, the verification of timestamp coverage, the requirement to treat the final interval as partial, and the distinction between traffic-volume observations and actual security conclusions.

The AI-generated interpretation identified the major temporal pattern as a high initial traffic burst, followed by a sharp reduction, followed by lower-volume activity with smaller fluctuations.

AI interpretation was not accepted as evidence by itself. The calculated timeline was subsequently checked directly against the actual Combined_Traffic_Data.csv. The independent verification confirmed that all 11 timeline interval counts displayed in the visualization exactly match the underlying CSV data. This demonstrates the use of AI as an analytical assistance and interpretation mechanism, while retaining analyst verification of the underlying statistical evidence.

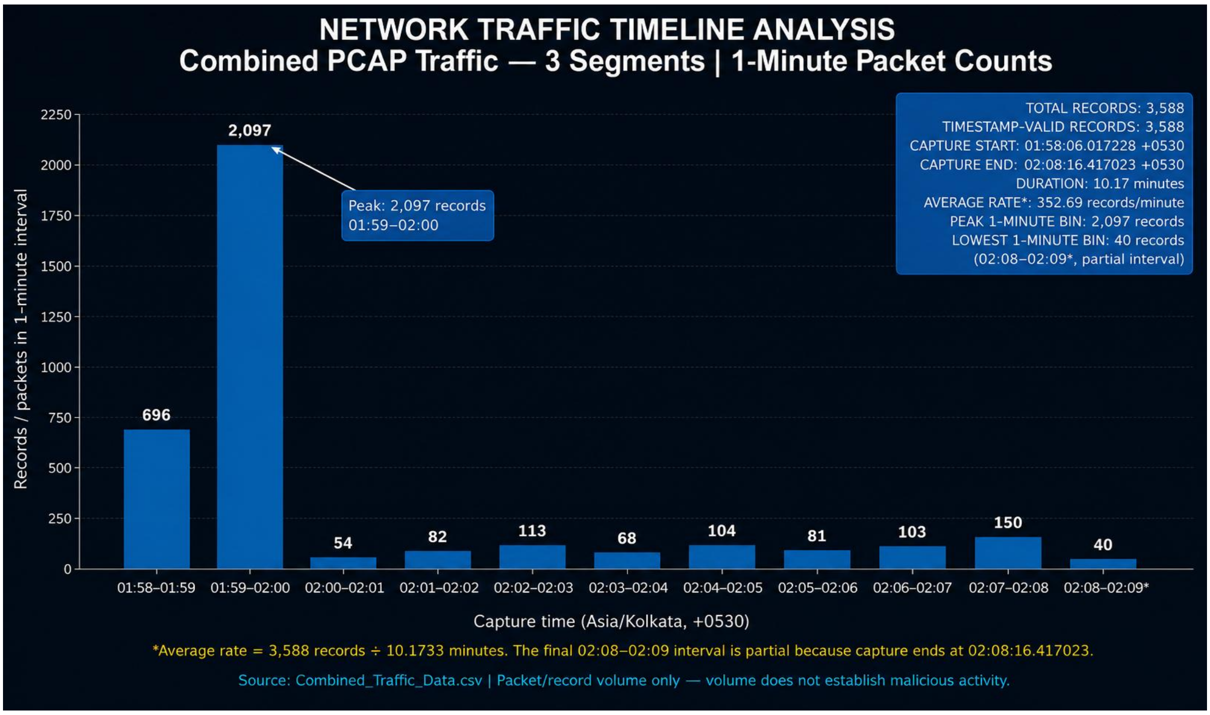
10. Evidence of Verification

The timeline analysis underwent an additional verification step after the visualization was generated. The actual combined CSV was programmatically processed independently of the visual output, confirming the following results.

Verification Item	Result
Total CSV records	3,588
Valid timestamp records	3,588
Timestamp validation exceptions	0
One-minute intervals	11
Timeline total	3,588
Peak interval	01:59 to 02:00
Peak count	2,097
Lowest interval	02:08 to 02:09 (partial)
Lowest count	40
Capture start	01:58:06.017228 +0530
Capture end	02:08:16.417023 +0530
Capture duration	approximately 10:10.40 (10.1733 minutes)
Interval values match visual	Yes, all 11

This verification establishes that the timeline visual represents the actual timestamp distribution of the combined dataset.

11. Timeline Visualization



Visual: Network Traffic Timeline Analysis

The visualization provides a graphical representation of the one-minute traffic distribution and highlights the 2,097-record peak during 01:59 to 02:00. It also identifies the partial final interval and provides the verified chronological distribution of all 3,588 records.

12. Security Interpretation

The timeline reveals an investigation-relevant traffic-volume anomaly, because a very large proportion of the captured records occurred within a single one-minute interval. The 01:59 to 02:00 interval contains 2,097 records, representing approximately 58.44% of all 3,588 records. This concentration warrants investigation because significant temporal bursts can be useful indicators when combined with other network evidence.

However, traffic volume alone does not establish malicious activity. The timeline cannot independently establish malware, command-and-control communication, beaconing, data exfiltration, endpoint compromise, attacker identity, or malicious intent.

Further contextual analysis would be required, including protocol analysis, endpoint/process investigation, DNS investigation, payload inspection where available, threat intelligence, and comparison with an appropriate network baseline. This conclusion remains consistent with the evidence limitations established in the Detection Rules analysis and the earlier traffic visualizations.

13. Relationship to Detection Rules Analysis

The timeline analysis provides supporting temporal evidence for the detection and investigation process developed earlier in Milestone 3. In particular, the timeline helps evaluate whether communication appears concentrated, repeated, bursty, or periodically distributed over time.

The dataset demonstrates a significant burst, particularly during 01:59 to 02:00, followed by substantially lower traffic volumes. However, the analysis did not establish a stable periodic interval. Therefore, the observed traffic burst should be treated as an investigation-relevant temporal observation, rather than being classified as confirmed beaconing.

This distinction is important because the detection rule mapping and overall security investigation dashboard explicitly state that detection alerts do not equal confirmed threats. The timeline therefore supports the broader investigation while maintaining the project's evidence-based security conclusions.

14. Final Conclusion

The combined PCAP dataset was successfully transformed into a verified chronological traffic timeline using timestamp-based one-minute aggregation.

The analysis confirmed that 3,588 total records were present in the combined dataset, and timestamp information was available for all 3,588 records used in the timeline analysis. All records were included in the chronological time binning. The capture lasted approximately 10 minutes 10.40 seconds (10.1733 minutes).

The highest traffic volume occurred during the 01:59 to 02:00 interval, with 2,097 records, representing approximately 58.44% of the complete 3,588-record dataset. Traffic volume decreased sharply immediately after the peak, and later traffic remained comparatively low, with smaller fluctuations. The final 02:08 to 02:09 interval contains 40 records but is partial because the capture ended at 02:08:16.417023.

All 11 timeline interval counts were independently verified against the Combined_Traffic_Data.csv, with the interval totals reconciling to all 3,588 records. CSV row order was not used to determine chronology; actual timestamp values were used instead.

The timeline therefore provides reliable statistical evidence of the temporal distribution and concentration of network traffic in the combined PCAP dataset. The most important security conclusion is that the large traffic burst is investigation-relevant but not, by itself, proof of malicious activity. Additional contextual and host-level evidence would be required before establishing malicious intent or compromise.